ACADEMIC PERFORMANCE & STUDENTS ANALYSIS

PROJECT OVERVIEW AND OBJECTIVE

OVERVIEW:

The Student Academic Analytics & Performance Dashboard is an end-to-end Power BI solution designed to analyze, visualize, and track key performance indicators (KPIs) across educational, demographic, and financial domains within an academic institution.

PROJECT OBJECTIVES:

To design an interactive Power BI dashboard for evaluating student academic performance, tracking demographic and geographical trends, monitoring attendance and library resource usage, and analyzing institutional fee collections and revenue streams.

DATA SOURCES

- **Source Description:** Excelx.com/practice-data/generators/education-academia
- **Timeline:** 01-08-2023 TO 02-11.2026.
- **Domain**: Education/ Academic Performance/ Department Enrolment / Library Engagement/ Institutional Financial/ Fees Collections & Students Analytics.

PROBLEM STATEMENT

- Track student performance and grade distribution across various academic departments.
- Analyze attendance percentages against assignment completion and pass/fail trends to spot early intervention opportunities.
- Monitor revenue streams, fee collection efficiency, and financial modes of payment.
- Limited Visibility into Student Geographical Distribution and Engagement.

ATTRIBUTE (COLUMN /FEATURES) DETAILS:

ATTRIBUTE NAME	DATA TYPE	DESCRIPTION
Department ID	Text (Varchar)	Unique identifier for each department (Primary Key, e.g., DEPT-101).
Department Name	Text (Varchar)	Full name of the academic department (e.g., Computer Science, Civil, Marketing).
School	Text (Varchar)	Parent academic faculty/school (e.g., Engineering, Business, Arts & Sciences).
Course ID	Text (Varchar)	Unique identifier for each course offered (Primary Key, e.g., 101-301).
Course Name	Text (Varchar)	Official title of the subject/course (e.g., Computer Science Fundamentals).
Department ID	Text (Varchar)	Foreign Key linking the course to its respective academic department.
Credits	Integer	Total academic credit points assigned to the

ATTRIBUTE NAME	DATA TYPE	DESCRIPTION
		course (e.g., 3, 4).
Assignment ID	Text (Varchar)	Unique identifier for each assignment (Primary Key, e.g., ASG-101-301-1).
Course ID	Text (Varchar)	Foreign Key linking the assignment to a specific course.
Title	Text (Varchar)	Name/description of the coursework (e.g., Assignment 1: CS Fundamentals).
Due Date	Date	Submission deadline for the coursework (e.g., 31 August 2023).
Enrollment ID	Text (Varchar)	Unique identifier for each student-course enrollment record (Primary Key).
Student ID	Text (Varchar)	Foreign Key linking to the enrolled student (e.g., STU-2023006).
Course ID	Text (Varchar)	Foreign Key specifying the enrolled course (e.g., 101-301).
Semester	Text (Varchar)	Academic term of enrolment (e.g., Fall, Spring).
Year	Integer	Academic year of enrolment (e.g., 2023).
Attendance ID	Text (Varchar)	Unique identifier for each daily attendance log (Primary Key).
Enrollment ID	Text (Varchar)	Foreign Key linking the attendance record to a specific student enrollment.
Date	Date	Specific date on which attendance was recorded.
Status	Text (Varchar)	Attendance status of the student on that date (e.g., Present, Absent).
Grade ID	Text (Varchar)	Unique identifier for each evaluation record (Primary Key).
Enrollment ID	Text (Varchar)	Foreign Key linking the grade to a specific enrollment record.
Grade	Text (Varchar / Char)	Letter grade awarded for course completion (e.g., A, A+, B, C, F).
Payment ID	Text (Varchar)	Unique transaction identifier (Primary Key, e.g., PAY-10).
Student ID	Text (Varchar)	Foreign Key identifying the student making the fee payment.
Amount	Decimal / Currency	Total monetary fee amount paid (e.g., 116652.09).
Payment Date	Date	Transaction completion date (e.g., 20 August 2023).
Semester	Text (Varchar)	Academic semester for which fees were paid (e.g., Fall).
Year	Integer	Academic year associated with the payment (e.g., 2023).
Method	Text (Varchar)	Payment channel used (e.g., Bank Transfer, Demand Draft, Online Gateway).
Loan ID	Text (Varchar)	Unique book borrowing transaction identifier (Primary Key).
Student ID	Text (Varchar)	Foreign Key linking to the borrower student.
Book Title	Text (Varchar)	Name of the borrowed library book (e.g., University Physics).

ATTRIBUTE NAME	DATA TYPE	DESCRIPTION
Loan Date	Date	Date on which the book was issued to the student.
Return Date	Date	Date on which the book was returned to the library.
Faculty ID	Text (Varchar)	Unique identifier for faculty members (Primary Key, e.g., FAC-201).
Faculty Name	Text (Varchar)	Full name of the faculty member (e.g., Dr. Pari Iyer).
Department ID	Text (Varchar)	Foreign Key associating faculty with their home department.
Rank	Text (Varchar)	Academic designation (e.g., Professor, Associate Professor, Lecturer).
Email	Text (Varchar)	Official university email address of the faculty member.
Student ID	Text (Varchar)	Unique institutional student identifier (Primary Key, e.g., STU-2023010).
Student Name	Text (Varchar)	Full name of the student (e.g., AYAAN SINGH).
Gender	Text (Varchar)	Gender identity of the student (e.g., Male, Female).
City	Text (Varchar)	Residential / origin city of the student (e.g., Pune).
Contact ID	Text (Varchar)	Unique identifier for the contact record (Primary Key).
Student ID	Text (Varchar)	Foreign Key linking to the specific student.
Email	Text (Varchar)	Student's official university email address.
Phone Number	Text (Numeric / Varchar)	Primary telephone contact number for the student.
Maximum Grade Points	Decimal / Integer	Numerical maximum score potential for academic evaluations.
Pass %	Decimal / Percentage	Calculated measure representing departmental student pass rate.
Avg_Assig_Score	Decimal	Average score achieved across assignment submissions.
Grade Status	Text (Varchar)	Categorical pass/fail or status evaluation indicator.

TOOLS & TECHNOLOGIES

- **Excel:** Data cleaning, transformation.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

DATA PRE-PROCESSING (EXCEL / POWER QUERY)

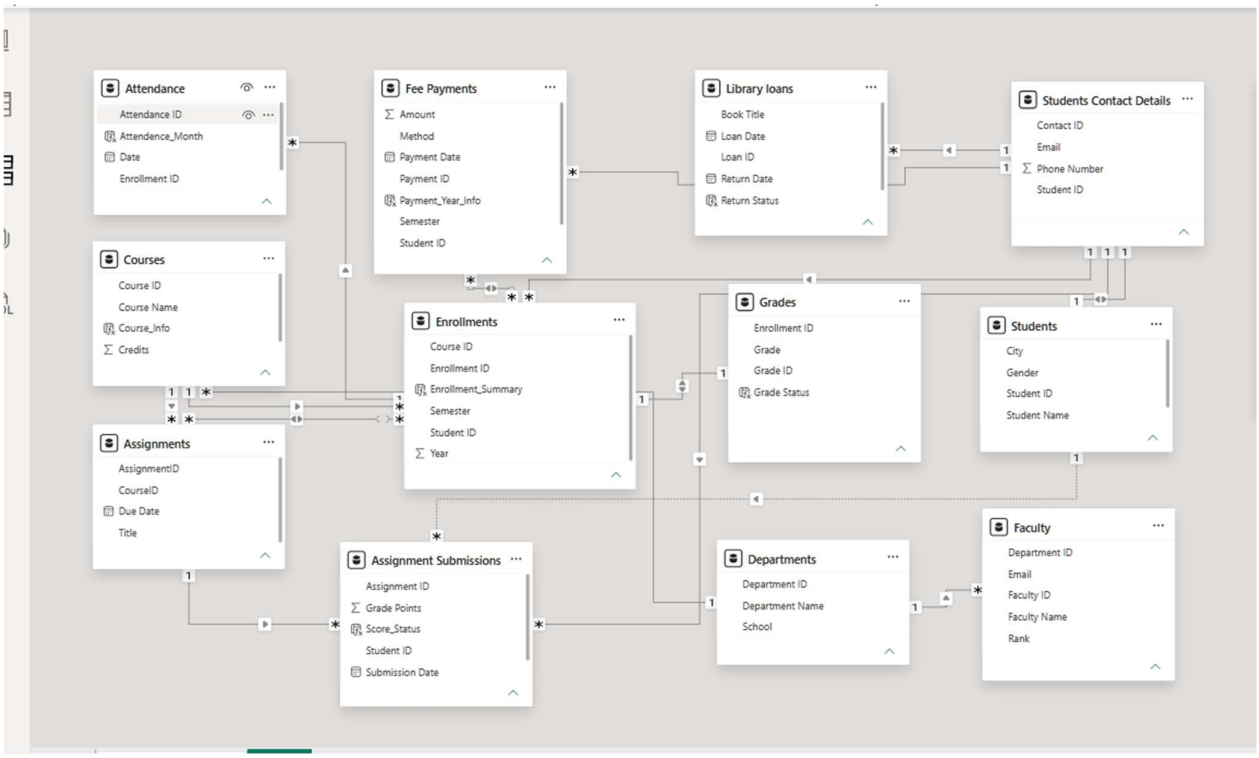
Tasks Performed:

- **Data Cleaning & Transformation:**
  - Data Consolidation & Integration (Raw tables to Consolidate worksheet).
  - Removing Duplicates (Duplicate Rows).
  - Handling Missing Values / Nulls (Missing data).
  - Data Type Standardization (Data types, Numeric, Text, Date Format).
  - Text Formatting & Cleaning (Extra Spaces, Upper, Lower & Inconsistence)
  - Star Schema – Dimension & Fact tables (Dimension tables, Performance metrics to Fact table).
  - Primary & Foreign Key Relationship (Data Integrity & relationship).

DATA MODELLING AND DAX (POWER BI):

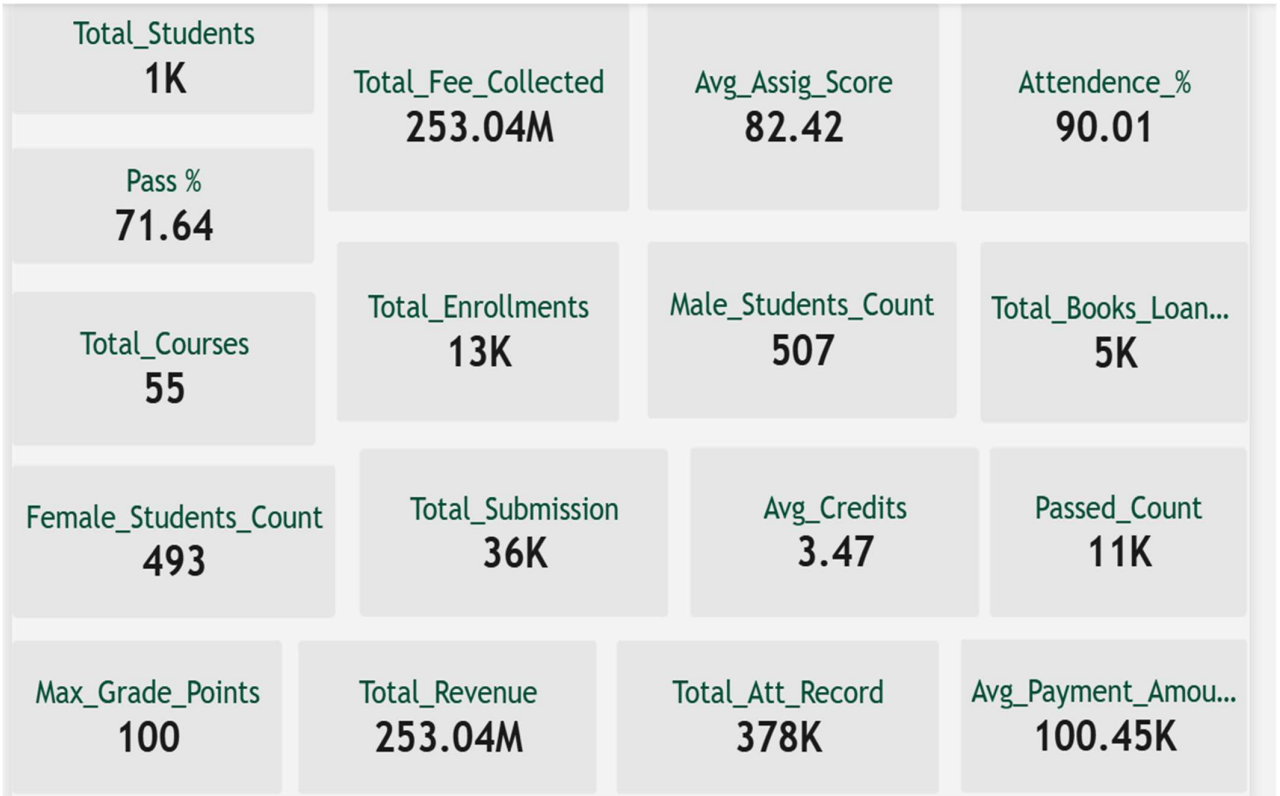
- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.

SCREENSHOT OF DATA MODELLING



- **Calculated Columns & DAX Measures:** Implemented DAX formulas for key metrics, such as total sales, growth percentages, and cumulative totals**. (List Out)**

S.NO	MEASURE NAME	SUGGESTED DAX FORMULA	VALUE
1	**Total_Students**	DISTINCTCOUNT (Students [Student ID])	1K
2	**Male_Students_Count**	CALCULATE (COUNT (Students [Student_ID]), Students [Gender] = "Male")	507
3	**Female_Students_Count**	CALCULATE (COUNT (Students [Student_ID]), Students [Gender] = "Female")	493
4	**Total_Courses**	COUNTROWS(Courses)	55
5	**Total_Enrollments**	COUNTROWS(Enrollments)	13K
6	**Total_Submission**	COUNTROWS(Submissions)	36K
7	**Total_Att_Record**	COUNTROWS(Attendance)	378K
8	**Total_Books_Loaned**	COUNTROWS(Book_Loans)	5K
9	**Passed_Count**	CALCULATE (COUNT (Grades [Grade_ID]), Grades [Status] = "Pass")	11K
10	**Pass %**	DIVIDE([Passed_Count], [Total_Enrollments], 0)	71.64%
11	**Avg_Assig_Score**	AVERAGE (Submissions [Score])	82.42
12	**Attendence_%**	DIVIDE (CALCULATE (COUNT (Attendance [Att_ID]), Attendance [Status] = "Present"), [Total_Att_Record], 0)	90.01%
13	**Avg_Credits**	AVERAGE (Courses [Credits])	3.47
14	**Max_Grade_Points**	MAX (Grades [Grade Points])	100
15	**Total_Fee_Collected**	SUM ('Fee Payments' [Amount])	253.04M
16	**Total_Revenue**	SUM (Payments [Amount])	253.04M
17	**Avg_Payment_Amount**	AVERAGE (Payments [Amount])	100.45K



ANALYSIS AND VISUALIZATIONS (POWER BI)

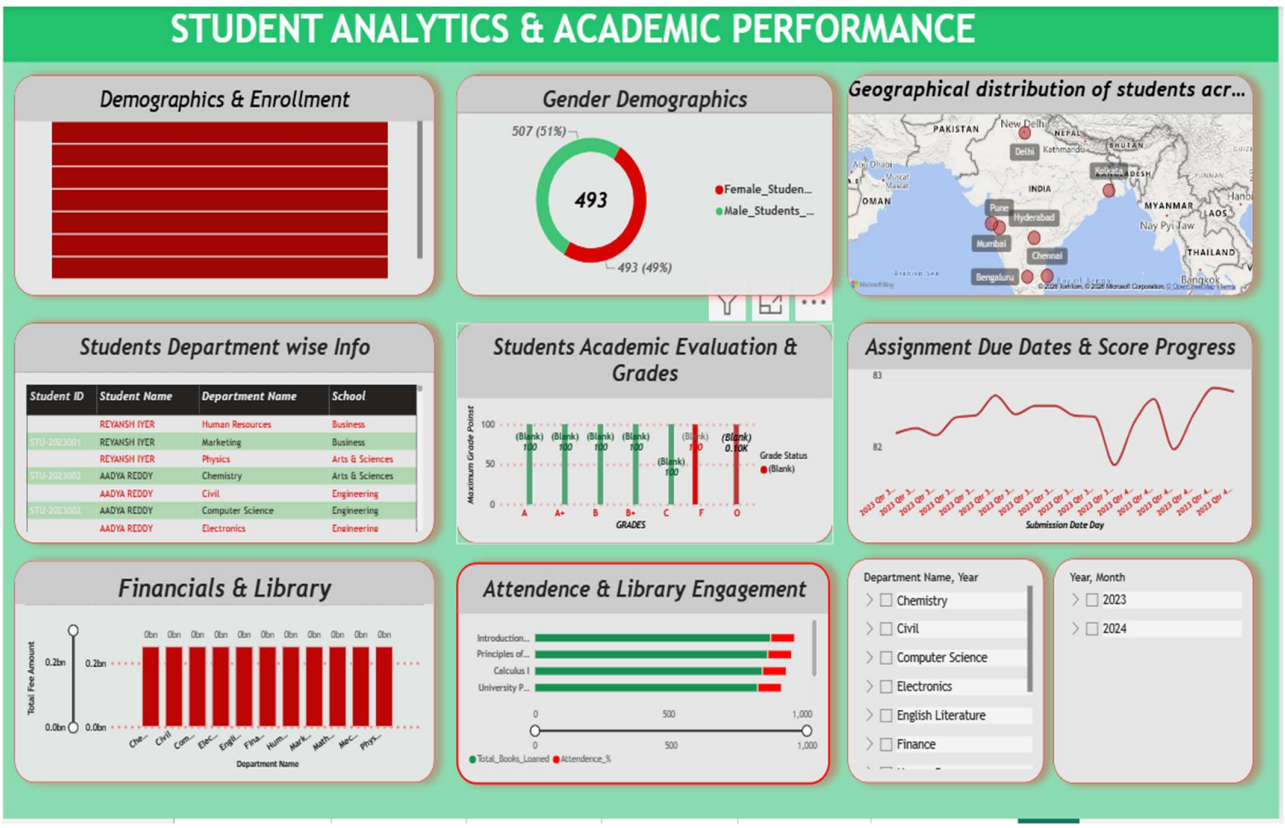
DASHBOARD FEATURES:

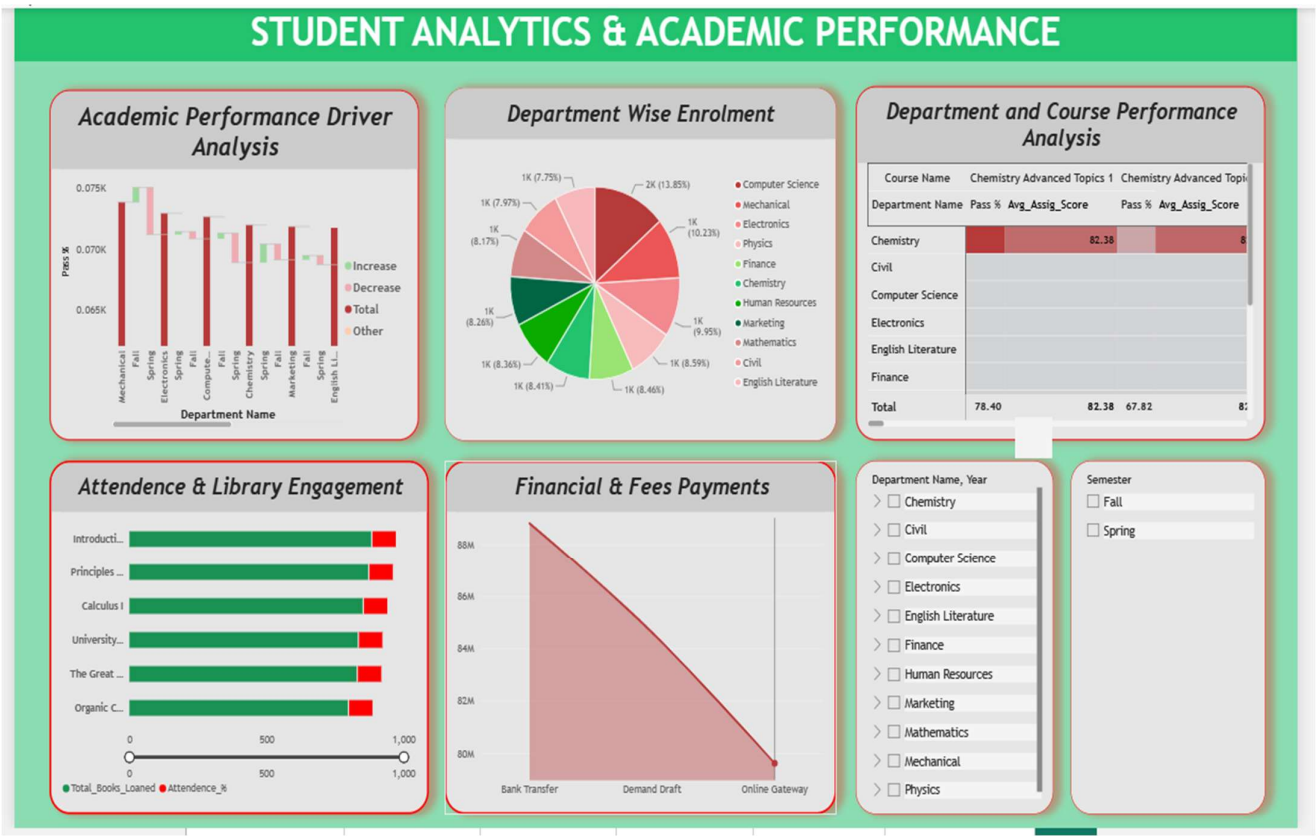
- **Academic Performance Driver Analysis (Waterfall Visual):** Evaluated how student pass percentages (Pass %) increased or decreased across departments and semesters using a waterfall chart.
- **Department Wise Enrolment (Donut Visual):** Displayed the percentage distribution of student enrollments across various departments such as Computer Science, Mechanical, Electronics, and Physics using a donut chart.
- **Department and Course Performance Analysis (Matrix Visual):** Compared department-specific course pass percentages (Pass %) and average assignment scores using a conditional-formatted matrix table.
- **Attendance & Library Engagement (Stacked Bar Chart):** Analyzed the relationship between the total number of books loaned from the library and student attendance percentages using stacked bar charts and slicers.
- **Financial & Fees Payments (Area / Trend Visual):** Analyzed fee collection trends across different payment methods including Bank Transfer, Demand Draft, and Online Gateway.
- **Gender Demographics (Donut Visual):** Broken down student gender ratios (Male: 507 / 51%, Female: 493 / 49%) clearly via a donut chart.
- **Geographical Distribution of Students (Map Visual):** Mapped student origins and geographical distribution across major cities (such as Chennai, Bengaluru, Mumbai, and Delhi) using a field map visual.
- **Students Department wise Info (Detailed Grid / Table Visual):** Created a detailed data grid displaying Student IDs, student names, department names, and school affiliations.
- **Students' Academic Evaluation & Grades (Lollipop / Bar Visual):** Compared student grade point distributions (Grades: A, A+, B, B+, C, F) and maximum grade points using bar visuals.
- **Assignment Due Dates & Score Progress (Line Chart Visual):** Highlighted assignment submission due dates and student score progression trends across quarters (Q1 to Q4) using a line chart.
- **Financials & Library Analysis (Column Chart Visual):** Illustrated total fee amounts collected per department using column visuals.

GENERAL INTERACTIVE FEATURES (INTERACTIVE SLICERS & FILTERS):

- **Dynamic Slicers:** Included interactive filters for Department Name, Year, Month, and Semester (Fall / Spring) to enable dynamic data exploration across both dashboard pages.

SCREENSHOT OF DASHBOARD /REPORT





INSIGHTS

KEY FINDINGS:

- **Total Students & Enrollment:**
    - The system tracks a total of 1K (1,000) students.
    - There are 55 total courses and 13K total enrollments.
    - Gender demographics show a balanced split with 507 (51%) Male students and 493 (49%) Female students.

- **Financials & Revenue:**
    - Total fee collection and revenue stand at $253.04M.
    - The average payment amount per transaction is $100.45K. Fees are processed through modes including Bank Transfer, Demand Draft, and Online Gateway.

- **Academic Evaluation & Attendance:**
    - The overall institutional pass percentage is 71.64\% with 11K passed student records.
    - The average assignment score across departments is 82.42.
    - Overall student attendance is high at 90.01\% out of 378\text{K} total attendance records.
    - Library engagement accounts for 5K total books loaned.

- **Department Analysis:**
    - Computer Science leads department enrollments at 13.85\% (2\text{K} enrollments).
    - Pass percentages and assignment scores vary across different departments and semesters (Fall vs. Spring).

- **Geographical & Engagement Insights:**
    - Students are distributed across major metropolitan hubs and cities across India (such as Bengaluru, Chennai, Mumbai, Hyderabad, and Delhi).
    - A positive correlation exists between library resource utilization (book loans) and student attendance and performance.
    - Summarize trends, patterns, or anomalies identified in the data.

PROVIDE THE ANALYSIS INSIGHTS:

DESCRIPTIVE ANALYSIS:

- **Total Student Metrics & Enrollment**: The system tracks a total student count of 1K (1,000 active students) distributed across 55 total courses and 13K total enrollments. Gender demographics reflect a balanced distribution with 507 (51%) Male students and 493 (49%) Female students.
- **Financial Performance:** Total fee collection and revenue stand at 253.04M, with an average payment amount of 100.45K per transaction. Fee payments are distributed across payment modes including Bank Transfer, Demand Draft, and Online Gateway.
- **Academic Evaluation & Attendance:** The overall institutional pass percentage is 71.64\% with 11K passed student records. The average assignment score across departments is 82.42, and overall student attendance is high at 90.01\% out of 378\text{K} total attendance records. Additionally, 5\text{K} total books are loaned through library engagement.
- **Geographical Distribution:** Students are broadly distributed across major regional hubs and metropolitan cities throughout India and neighboring regions (such as Bengaluru, Chennai, Mumbai, Hyderabad, Delhi, and others).

DIAGNOSTIC ANALYSIS:

- **Departmental Variance in Performance:** The Academic Performance Driver Analysis and Department and Course Performance Analysis reveal that specific departments experience fluctuations in pass percentages and assignment scores between semesters (Fall vs. Spring). For instance, core technical departments like Computer Science, Mechanical, and Civil show distinct grade distributions and pass rate drivers influenced by assignment regularity.
- **Grade Distribution Bottlenecks:** The Students Academic Evaluation & Grades visual indicates that while the majority of students score within passing grade brackets (A, A+, B, B+, C), a noticeable cluster falls into failing grades (F) or unassigned/blank status categories due to missed submissions or lower maximum grade points.
- **Engagement Correlation:** The Attendance & Library Engagement chart demonstrates a direct positive correlation between student attendance percentages and total books loaned per course (e.g., Introduction to AI, Calculus I, University Physics, Organic Chemistry), showing that students with higher library resource utilization consistently maintain better attendance and scores.

PREDICTIVE ANALYSIS:

- **Submission Trends & Score Forecasting:** Based on the Assignment Due Dates & Score Progress time-series trend line, assignment scores fluctuate cyclically between quarters (e.g., Q3 and Q4 2023/2024). Projecting these trends indicates that upcoming assignment deadlines in subsequent academic terms may

experience similar dip-and-recovery patterns around evaluation windows unless proactive submission reminders are triggered.

- **Enrollment & Revenue Projections:** Given the current distribution in Department Wise Enrolment (where Computer Science leads at 13.85\% / 2\text{K} enrollments, followed by other core branches), future fee revenues are projected to scale proportionately with high-demand technical courses.
- **At-Risk Student Identification:** Based on the current institutional pass rate of 71.64\%, roughly 28.36\% of students are statistically susceptible to underperforming or failing upcoming semester evaluations if attendance drops below the 90\% threshold.

PRESCRIPTIVE ANALYSIS:

- **Targeted Academic Interventions:** Implement mandatory remedial coaching and faculty check-ins for departments exhibiting lower pass percentages (such as specific advanced topics in Chemistry and foundational engineering modules) to lift the overall institutional pass rate above the current 71.64\% mark.
- **Engagement & Resource Optimization:** Leverage the positive correlation between library engagement and attendance by incentivizing digital and physical book loans for courses showing lower student participation (e.g., Principles of Marketing, The Great Gatsby).
- **Automated Submission Tracking:** Utilize the Assignment Due Dates & Score Progress insights to set up automated Power BI alerts or notifications for students approaching due date troughs, ensuring assignment submission rates exceed the current 36\text{K} total volume baseline.
- **Financial Process Streamlining:** Since payment collections span multiple channels (Bank Transfer, Demand Draft, Online Gateway), promote automated digital gateways to reduce transaction friction and optimize the 253.04M total revenue collection pipeline.

CONCLUSIONS:

The Education Performance Overview Power BI dashboard successfully synthesizes institutional data across 1K students, 13K enrollments, and 55 courses to deliver actionable insights into academic performance, demographics, and finances. Featuring a strong 90.01\% attendance rate and a 71.64\% overall pass rate alongside 253.04M in total fee collection, the dashboard highlights key operational successes while surfacing departmental variations and library engagement trends. By combining descriptive metrics with predictive and prescriptive analysis, this analytics tool equips academic leadership with the strategic visibility needed to boost student outcomes, streamline payment collections, and optimize overall institutional performance.
